A covariant finite-density exchange model: conservation, canonical density dynamics, and limits of nonlinear force matching

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Interacting cosmologies often postulate a transfer vector or infer a local force normalization from homogeneous evolution without deriving both from a common action. We study a conditional four-dimensional model containing Einstein gravity, pressureless matter coupled through a conformal metric, a complex finite-charge scalar, an independent force field and timelike frame, and a real reservoir. Direct variation yields finite interface currents

$$\nabla_\mu T_m^{\mu\nu} = Q^\nu_{\rm mp}, \quad \nabla_\mu T_P^{\mu\nu} = -Q^\nu_{\rm mp} + Q^\nu_{\rm syn}, \quad \nabla_\mu T_R^{\mu\nu} = -Q^\nu_{\rm syn},$$

whose sum vanishes on shell. A registered homogeneous solution exhibits nonzero reversible exchange while conserving the full scalar charge. On a formal reduced slow branch, the physical dust-density coordinate has $K_D = a^5\rho_m/k^2 > 0$, exactly matching the uncoupled normalization of an independent coupled-fluid effective field theory. This does not establish full stability: the density Hamiltonian is indefinite in its growing-mass direction, and an equal-Newton-coupling family retains a defective double zero-speed scalar root.

The homogeneous equations are insensitive to the cubic spatial-force coefficient, whose invariant normalization is therefore not identifiable from background matching. On compact T^3, the static force equation also requires a zero-mean source; a universal pointwise square-root prescription fails a curl test, and a finite fourth-derivative regulator makes the arbitrarily weak fixed-scale response linear rather than square-root. These are scoped obstructions, not a rejection of every nonlinear regime. The model provides an action-derived source-vector construction and sharp consistency tests, but not a completed cosmology, a derived MOND normalization, or a physics-stable perturbation theory.

I. INTRODUCTION

Three questions are often conflated in interacting cosmology:

1. Is the exchange current derived from a complete off-shell action?
2. Is the resulting constrained dynamical system stable and well posed?
3. Does the model reproduce a proposed weak-field or cosmological phenomenology?

An affirmative answer to the first does not answer the other two. This paper keeps that separation explicit.

We present the R4C1 candidate as a conditional construction. Its matter coupling is a concrete pressureless-fluid realization inspired by covariant fluid actions [1]; its normalized timelike-frame sector is related in form to Einstein–aether models [2]; and its cubic spatial force operator has the same nonrelativistic variational structure as the deep-AQUAL term [3]. These precedents do not select the R4C1 coefficients or establish its physical viability.

The positive result is an action-level source vector with nonsingular zero-coupling limits, a finite-charge background control, and a canonically positive dust-density kinetic coefficient. The equally important negative results are that homogeneous evolution does not identify the nonlinear force coefficient, compact periodicity forbids a static positive-mean Poisson source, the finite regulator changes the weak-source asymptote, and the current equal-Newton family has an unresolved defective scalar zero branch.

All results below are conditional on the declared action, chart, boundary conditions, and registered domains. No observational data are used to select its coefficients.

II. FIELDS, DOMAIN, AND CONDITIONAL ACTION

Use signature $(-+++)$, $c = \hbar = 1$, and spacetime $\mathcal{M} = I \times T^3$. The independent fields are

$$g_{\mu\nu},\ U^\mu,\ \lambda_U,\ u,\ v,\ \psi,\ r,\ \tau,\ \epsilon,\ z.$$

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Repository: `github.com/brendohxd/ITSM-...`

Write $\Phi = (u+iv)/\sqrt{2}$, $s = u^2+v^2$, and define

$$\ell = \sqrt{-g_{\mu\nu}U^\mu U^\nu}, \qquad n^\mu = U^\mu/\ell, \tag{1}$$

$$h^{\mu\nu} = g^{\mu\nu} + n^\mu n^\nu, \qquad J_\mu = v\nabla_\mu u - u\nabla_\mu v, \tag{2}$$

$$Q = n^\mu \nabla_\mu \psi, \qquad Y = h^{\mu\nu}\nabla_\mu\psi\nabla_\nu\psi, \tag{3}$$

$$a^\mu = n^\nu \nabla_\nu n^\mu, \qquad \theta = \nabla_\mu n^\mu, \tag{4}$$

$$\Delta\psi = \nabla_\mu(h^{\mu\nu}\nabla_\nu\psi) - a^\mu\nabla_\mu\psi. \tag{5}$$

The last expression equals $h^{\mu\nu}\nabla_\mu\nabla_\nu\psi + \theta Q$. Normalization is performed off shell. The chart is restricted to future-timelike U^μ with $\ell > 0$; no division by s or a conserved charge is used.

The action is

$$S = \int_{\mathcal{M}} \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\mathrm{P}}^2}{2}R - \rho_\Lambda + \mathcal{L}_P + \mathcal{L}_R + \mathcal{L}_m\right] + S_{\mathrm{GHY}}, \tag{6}$$

with

$$V(s) = \frac{m^2 s}{2} + \frac{\lambda_4 s^2}{8} + \frac{\lambda_6 s^3}{24\Lambda^2}, \tag{7}$$

$$V_R(r) = \frac{m_r^2 r^2}{2} + \frac{\lambda_r r^4}{4}, \tag{8}$$

$$W(s,r) = \frac{g_r s r^2}{4}, \tag{9}$$

$$\mathcal{L}_P = -\frac{1}{2}(\nabla u)^2 - \frac{1}{2}(\nabla v)^2 - V - W + \mathcal{L}_U + \mathcal{L}_{\mathrm{align}} + \mathcal{L}_{\mathrm{force}}, \tag{10}$$

$$\mathcal{L}_R = -\frac{1}{2}(\nabla r)^2 - V_R(r), \tag{11}$$

$$\mathcal{L}_m = -\frac{\epsilon}{2}\left[C^2(\nabla\tau)^2 + C^4\right], \tag{12}$$

$$C(\psi) = e^{\beta\psi}. \tag{13}$$

Matter therefore follows the conformal metric $\widetilde{g}_{\mu\nu} = C^2 g_{\mu\nu}$.

The frame, alignment, and force blocks are

$$\mathcal{L}_U = -\frac{M_U^2}{2}\big[c_1(\nabla_\mu U_\nu)(\nabla^\mu U^\nu) + c_2(\nabla_\mu U^\mu)^2 + c_3(\nabla_\mu U_\nu)(\nabla^\nu U^\mu) - c_4(U^\rho\nabla_\rho U_\mu)(U^\sigma\nabla_\sigma U^\mu)\big] + \frac{\lambda_U}{2}(U^2+1), \tag{14}$$

$$\mathcal{L}_{\mathrm{align}} = -\frac{\zeta}{2}h^{\mu\nu}J_\mu J_\nu, \tag{15}$$

$$\mathcal{L}_{\mathrm{force}} = \frac{K_Q}{2}Q^2 - AY^{3/2} + \frac{b}{2}z^2 - bh^{\mu\nu}\nabla_\mu z\nabla_\nu\psi - bz\,a^\mu\nabla_\mu\psi. \tag{16}$$

For $b > 0$, the auxiliary equation is $z = -\Delta\psi$, yielding $-b(\Delta\psi)^2/2$ in the bulk with the frozen boundary convention. At $b = 0$ the auxiliary is omitted; dividing its equation by b would incorrectly preserve a rank-changing constraint.

Positive K_Q, A, b, ζ define the exploratory interior. These are declared coefficients, not first-principles values. The c_i are likewise not certified stable.

III. VARIATION, CHARGE, AND SOURCE-VECTOR CLOSURE

The force momenta and equations are

$$\Pi_\psi^\mu = K_Q Q n^\mu - 3A\sqrt{Y}\,h^{\mu\nu}\nabla_\nu\psi - bh^{\mu\nu}\nabla_\nu z - bza^\mu, \tag{17}$$

$$E_z = b(z + \Delta\psi), \tag{18}$$

$$E_\psi = -\nabla_\mu\Pi_\psi^\mu + \beta T_m. \tag{19}$$

The dust variations give

$$\nabla_\mu(\epsilon C^2\nabla^\mu\tau) = 0, \qquad C^2(\nabla\tau)^2 + C^4 = 0. \tag{20}$$

The physical dust stress is obtained only after varying the off-shell action:

$$T_{\mu\nu}^m = \epsilon C^2\nabla_\mu\tau\nabla_\nu\tau + g_{\mu\nu}\mathcal{L}_m, \qquad T_m = -\epsilon C^4 \tag{21}$$

on the multiplier constraint.

The global $U(1)$ transformation of u, v yields

$$j^\mu = J^\mu + \zeta s h^{\mu\nu}J_\nu, \qquad \nabla_\mu j^\mu = vE_u - uE_v. \tag{22}$$

Thus the full charge, not the bare phase current alone, is conserved on the condensate equations. Energy exchange does not imply charge creation.

Direct stress variation gives, in the baseline split,

$$Q_{\mathrm{mp}}^\nu = \beta T_m\nabla^\nu\psi, \tag{23}$$

$$Q_{\mathrm{syn}}^\nu = -W_r\nabla^\nu r = -\frac{g_r}{2}sr\nabla^\nu r, \tag{24}$$

and the on-shell sector identities

$$\nabla_\mu T_m^{\mu\nu} = Q_{\mathrm{mp}}^\nu, \tag{25}$$

$$\nabla_\mu T_P^{\mu\nu} = -Q_{\mathrm{mp}}^\nu + Q_{\mathrm{syn}}^\nu, \tag{26}$$

$$\nabla_\mu T_R^{\mu\nu} = -Q_{\mathrm{syn}}^\nu. \tag{27}$$

Their sum vanishes. Both currents are polynomial and finite at $s = 0$ and at zero coupling. Reassigning a fraction of W between named sectors changes Q_{syn}^ν by a gradient while leaving the total action unchanged; the subsystem split must therefore be fixed before treating a named transfer current as observable.

The exact algebraic GR endpoint sets $\beta = g_r = M_U^2 = \zeta = K_Q = A = b = 0$, $u = v = r = 0$, constant ψ, and $\lambda_U = 0$, while omitting z. Merely setting the two exchange couplings to zero is not pure GR because the remaining sectors still gravitate.

IV. HOMOGENEOUS FINITE-CHARGE BACKGROUND

For an FLRW metric and homogeneous fields, the currents reduce to

$$Q^0_{\rm mp} = \beta\rho_m\dot{\psi}, \qquad Q^0_{\rm syn} = \frac{g_r}{2}sr\dot{r}, \tag{28}$$

and

$$\dot{\rho}_m + 3H\rho_m = Q^0_{\rm mp}, \tag{29}$$

$$\dot{\rho}_P + 3H(\rho_P + p_P) = -Q^0_{\rm mp} + Q^0_{\rm syn}, \tag{30}$$

$$\dot{\rho}_R + 3H(\rho_R + p_R) = -Q^0_{\rm syn}. \tag{31}$$

The homogeneous alignment term vanishes and $a^3s\dot{\Theta}$ is the conserved charge in the nonzero-amplitude phase chart.

A registered finite-interval background was solved by independent DOP853 and Radau methods. It maintained finite positive a, H, ρ_m, nonzero interface currents, charge drift below 1.9×10^{-11}, dust-integral drift below 2.9×10^{-12}, and a finest independently differentiated sector-balance residual of 7.96×10^{-8}. The reservoir transfer changed sign. These are checks of a conditional finite-charge solution in arbitrary reference units; they are not measurements of H_0, cosmological abundances, entropy production, or global existence.

The spatial cubic term vanishes when $Y = 0$. Hence the background solution, charge, and interface currents do not determine A. This fact drives the identifiability result in Sec. VI.

V. CANONICAL DENSITY REDUCTION AND AN INDEPENDENT EFT CONTROL

On the registered formal slow branch, let D be the physical leading dust density contrast at nonzero comoving wavenumber k. A time-dependent canonical transformation from the inherited phase coordinate gives

$$L_D = \frac{K_D}{2}\left(\dot{D}^2 - M_D D^2\right), \tag{32}$$

$$K_D = \frac{a^5\rho_m}{k^2} > 0, \qquad M_D = -\frac{\rho_m}{2M_c^2}. \tag{33}$$

This result holds for $a, \rho_m, k > 0$ on the declared regular chart. The $k = 0$, $a = 0$, $\rho_m = 0$, and rank-changing auxiliary surfaces are not inverted.

The conjugate momentum and Hamiltonian are

$$P_D = K_D\dot{D}, \qquad H_D = \frac{P_D^2}{2K_D} + \frac{K_D M_D D^2}{2}. \tag{34}$$

The kinetic weight is positive. Because $M_D < 0$, however, the Hessian in (P_D, D) is indefinite. This is the familiar growing dust-density direction, not a negative dust kinetic coefficient; nor is it a proof that the full coupled Hamiltonian is bounded.

A useful independent convention check follows from the coupled-fluid EFT of Aoki et al. [4]. Their scalar variable is $X^T = (\delta_c/k, \zeta, \delta_m/k)$, their quadratic action has no overall factor $1/2$, and

$$K_{11} = \frac{a^2M^2H^2}{2}\left(3\Omega_c + \alpha_{m2} - \bar{\alpha}_{m1}^2\right). \tag{35}$$

Converting to $\frac{1}{2}K_D\dot{\delta}_c^2$ gives

$$K_D^{\rm EFT} = \frac{2a^3K_{11}}{k^2}. \tag{36}$$

In the uncoupled-dust limit, $3\Omega_c = \rho_c/(M^2H^2)$, and therefore

$$K_D^{\rm EFT} = \frac{a^5\rho_c}{k^2}, \tag{37}$$

exactly matching Eq. (33). This is a nontrivial normalization control. It does not identify the R4C1 interaction with α_{m2} or $\bar{\alpha}_{m1}$, and it does not import the EFT's small-scale stability conditions into R4C1.

VI. WHAT HOMOGENEOUS MATCHING CANNOT IDENTIFY

Under the positive field-chart change $\psi_{\rm new} = \sigma\psi$, $z_{\rm new} = \sigma z$,

$$K_{Q,\rm new} = K_Q/\sigma^2, \qquad A_{\rm new} = A/\sigma^3, \tag{38}$$

$$b_{\rm new} = b/\sigma^2, \qquad \beta_{\rm new} = \beta/\sigma. \tag{39}$$

The physical coefficient combinations are therefore

$$\frac{A}{K_Q^{3/2}}, \qquad \frac{b}{K_Q}, \qquad \frac{\beta}{\sqrt{K_Q}}. \tag{40}$$

Changing A alone changes the first invariant and is not a chart renaming. Nevertheless A is absent from the homogeneous equations because $Y = 0$. Background matching cannot select it.

The local static frozen-frame reduction is

$$3A\nabla\cdot(|\nabla\psi|\nabla\psi) - b\Delta^2\psi = \beta\rho. \tag{41}$$

For a regulator-subleading spherical exterior one obtains only the conditional leading expression

$$\psi'(R) = \frac{1}{R}\sqrt{\frac{\beta M}{12\pi A}}, \qquad a_{\rm dyn} \equiv \frac{g_\psi^2}{g_{\rm bar}} = \frac{\beta^3}{12\pi G_{\rm static}A}. \tag{42}$$

The required regulator ratio is $2b/(3ADR)$, with $D = \sqrt{\beta M/(12\pi A)}$; no common domain proving all local, quasistatic, frame, and metric approximations has yet been established. Equation (42) shows directly that a MOND-like normalization remains a free action-level combination. It does not derive a_0, a projection factor, or the historical value $2/3$.

The cubic term is C^2 but not C^3 at zero spatial gradient. Its gradient and Hessian vanish there, so regular linearization about $Y = 0$ cannot recover A. Nonlinear backgrounds can depend on it, but that is a different calculation.

VII. COMPACT-DOMAIN COMPATIBILITY CONDITIONS

A. Zero mode

On aligned flat FLRW, the exact force equation is

$$-K_Q a^{-3}\partial_t(a^3\dot{\psi}) + 3Aa^{-3}\nabla_x\cdot(|\nabla_x\psi|\nabla_x\psi) - ba^{-4}\Delta_x^2\psi - \beta\rho = 0. \quad (43)$$

Both spatial divergences integrate to zero on T^3, so

$$K_Q a^{-3}\partial_t(a^3\dot{\bar{\psi}}) = -\beta\bar{\rho}. \quad (44)$$

A positive mean density is incompatible with a fully static scalar in this reduction. The mean source must be evolved, subtracted as a background, compensated by another sector, or treated by a different equation. It cannot be inserted into a periodic Poisson problem.

After subtracting the mean and freezing a declared static snapshot,

$$3A\nabla\cdot(|\nabla\psi|\nabla\psi) - b\Delta^2\psi = \beta\,\delta\rho, \qquad \overline{\delta\rho} = \bar{\psi} = 0. \quad (45)$$

A three-dimensional pseudospectral control on odd grids $17, 25, 33$ passed its registered residual and convergence checks. Omitting b at the nonlinear solution left a normalized residual near 0.919, showing that the regulator is quantitatively important for that probe. The control is not a physical galaxy solution because metric, frame, dust, and Euler responses were frozen.

B. Curl obstruction to a pointwise square-root law

A force derived from a scalar must be curl free. For a generic Newtonian potential, the pointwise prescription

$$\mathbf{v} = \kappa\,\frac{\nabla\Phi_N}{|\nabla\Phi_N|^{1/2}} \quad (46)$$

has nonzero curl even though $|\mathbf{v}|\mathbf{v} = \kappa^2\nabla\Phi_N$. An explicit periodic counterexample is $\Phi_N = \varepsilon(\cos x + 2\cos y)$. Therefore the universal pointwise assignment cannot equal $\nabla\psi$ even locally on every allowed source. This does not reject nonlocal projected solutions, special symmetry classes, or a PDE solved with its harmonic and curl components retained.

C. Finite-regulator weak-source limit

For fixed nonzero b, fixed source shape, and contrast amplitude $\varepsilon \to 0$, the periodic minimizer obeys

$$\psi_\varepsilon = \varepsilon\psi_1 + O(\varepsilon^2) \quad \text{in } H^2(T^3), \quad (47)$$

with a linear biharmonic leading response. Hence the force is $O(\varepsilon)$, whereas a fixed positive square-root correction would be $O(\sqrt{\varepsilon})$. The latter cannot be the arbitrarily weak, fixed-scale asymptote of Eq. (45). A finite intermediate nonlinear window is not excluded; it must be derived with an EFT cutoff and the omitted coupled response.

D. Topology-sector quantum control

The separate rectangular-T^3 Casimir programme finds that the nonzero-mode conformal-scalar stress agrees with an independent heat-kernel/Mellin implementation and with the formula of Negro et al. [5]. That result is a useful topology-sector control, not a closure of R4C1. The state-dependent isotropic zero mode and renormalized dynamical backreaction remain separate problems.

VIII. GR APPROACHES AND THE STABILITY LEDGER

Turning off exchange at the original B1 coefficients does not recover equal cosmological and static gravitational couplings:

$$\frac{G_{\rm cos}}{G_{\rm static}} = \frac{5}{6}. \quad (48)$$

The exact algebraic endpoint of Sec. III does recover Einstein–dust, but it need not be a healthy continuous limit of every finite-density branch.

A separately frozen one-parameter family with $0 < \eta \leq 1$ enforces $G_{\rm cos} = G_{\rm static}$ and has

$$c_T^2 = \frac{8}{8-\eta}, \qquad K_{\rm vector} = \frac{\eta}{6}, \qquad M_c^2 = 1 - \frac{\eta}{12}. \quad (49)$$

Its background approaches Einstein–dust over the tested finite interval as $\eta \to 0$. This is a distinct path, not a repair of B1.

After the registered scalar constraint reduction, the finite-η family has positive canonical kinetic weights and nonzero leading speed squares

$$1, \quad 1, \quad 1 + \frac{\eta(u^2+v^2)}{13}, \quad \frac{4}{3(8-\eta)}. \quad (50)$$

However, its principal generator also has a double zero-speed root with geometric multiplicity one. The resulting Jordan chain is defective for every registered finite η. The family therefore does not establish strong hyperbolicity or a healthy physical EFT.

The correct present classification is therefore *conditional action with substantive physical holds.* Check counts are reproducibility evidence, not a physics pass.

IX. THERMODYNAMIC AND PHENOMENOLOGICAL BOUNDARIES

The reservoir field in Eq. (6) is reversible. No temperature, chemical potential, particle-production rate,

TABLE I. Current stability and closure ledger. Local checks and exact algebra do not override the open physical items.

Item	Current result	Boundary
Uncoupled dust-density kinetic normalization	$K_D = a^5\rho_m/k^2 > 0$	Nonzero k, a, ρ_m; formal slow branch
Density Hamiltonian	Indefinite growing-mass direction	Not a negative kinetic sign; no all-sector energy theorem
Tensor and vector sectors in equal-Newton family	Positive listed weights, real registered speeds	Vacuum/registered background controls, not PPN or interacting-domain proof
Nonzero scalar principal roots	Real with positive squared speeds	Formal high-momentum symbol only
Scalar zero branch	Defective double root	Strong hyperbolicity and subleading evolution unresolved
Fourth-spatial-derivative force branch	Formal p^4 fast mode	No physical cutoff or scattering window established
Constraint singularities	Excluded rather than inverted	$k = 0$, $H = 0$, zero-density and rank-changing endpoints need separate charts
Quantum and renormalized stability	Open	No all-sector determinant or state-selected semiclassical action

TABLE II. Falsifiers for the conditional branch.

Claim route	Reject or suspend if
Source-vector closure	Direct variation fails total on-shell conservation or a current becomes singular on the declared finite-field domain
Physical scalar EFT	A reduced ghost, nonreal characteristic, unresolved defective principal block, or nonunique evolution persists on the claimed domain
Weak-field square-root regime	No common regulator-subleading, quasistatic, periodic and matter-frame domain exists
GR recovery	No continuous finite-density path reaches the required local, cosmological and wave-sector GR observables
Cosmological application	One common parameter set fails background, perturbation, BBN or local-gravity likelihoods

dissipative response matrix, or coarse-grained entropy current is specified. The exchange currents therefore do not establish positive entropy production or a nonzero condensate-number source. A companion entropy-accounting manuscript derives the additional constitutive data required for such a claim.

Likewise, the current action does not yet provide a validated map to $H(z)$, $G_{\rm eff}(k, z)$, gravitational slip, BBN, a lensing potential, or a universal galactic force law. Table II states the near-term failure conditions.

X. DISCUSSION AND CONCLUSIONS

R4C1 demonstrates that the source-vector problem can be addressed without postulating Q^ν by hand. A complete declared action yields finite matter–plenum and plenum–reservoir currents, total on-shell conservation, a finite-charge background, and the standard uncoupled dust-density kinetic normalization.

The same calculation narrows the physical claim. The background cannot identify the nonlinear spatial-force coefficient; the compact zero mode forbids a static positive-mean source in the frozen reduction; the universal pointwise square-root shortcut fails an integrability test; and finite b changes the arbitrarily weak-source scaling. The positive density kinetic coefficient does not remove the indefinite growing direction or the defective zero-speed scalar branch.

Accordingly, this paper presents a conditional action, conservation theorem, normalization control, and scoped obstructions. It does not claim that ITSM is established, that a_0 or a 2/3 coefficient has been derived, that all perturbations are stable, or that cosmological data prefer this branch. Deferred independent review remains required for canonical promotion, while the scientific holds remain substantive irrespective of review status.

CODE, DATA, AND PROVENANCE

Contracts, derivations, executables, and immutable JSON receipts are available in the companion repository. Principal source SHA-256 prefixes are: action freeze `81bfc2e4f17b`, first variation `4834ea7517fd`, interact-

ing background `48b49fddd906`, coefficient identifiability `c4af4084ae83`, density action `8a1c2245fc02`, equal-Newton family `1b9fbc4185b6`, and equal-Newton scalar analysis `4d2ea777b8ed`. Hashes establish artifact identity, not independent physical validation.

DECLARATION ON THE USE OF ARTIFICIAL-INTELLIGENCE TOOLS

Artificial-intelligence-assisted tools were used for editorial drafting, source-code assistance, and checks of symbolic and numerical consistency. The author independently reviewed the manuscript, equations, cited sources, and generated artifacts and accepts full responsibility for the content.

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